

MAGNETOM

Safety-Related Tests Report System

Standard Procedure according to IEC 62353

Customer:	
Address:	
Fax / E-Mail:	
Fax / E-Mail:	

Department:		Room:	
Material-No.:		Serial-No.:	
Contract-No.:		Expire date:	
Order-No.:		System - ID:	

Avanto

Espreo

The instructions M6-020.833.05.03.XX are required for this protocol

Final Evaluation

☐ Initial Measurement ☐ Repeat Measurement

The evaluation was performed after completing all work steps.

- ☐ No deficiencies
- ☐ Slight deficiencies that require corrective measures: These must be recorded under Remarks.
- ☐ Significant deficiencies: The safety of the system is no longer assured. If these deficiencies are not corrected, there is a **"risk for patients, users or third parties"**.

Signature: _____

Date:

Tester:

If required by country-specific regulations:

The customer or a representative has taken note of the result of the evaluation of the system condition.

Signature: _____

Date:

Name:

Remarks:

Remarks regarding the Protocol

The chapter numbers refer to the chapters in the instruction, which is referenced on the cover page.

All pages have to have the serial number of the system and the date of maintenance in the page header.

The assignment n.a. (not applicable) indicates that the checkpoint or measured value is not used for this system.

On page 2 the completeness and the results of the maintenance work is confirmed.

Explanation of Abbreviations in the Report

Abbrev.	Explanation		Abbrev.	Explanation
SI	Safety Inspection		SIE	Electrical Safety Inspection
SIM	Mechanical Safety Inspection		CSE	Customer Service Engineer

Measuring Equipment and Measuring Instruments Used

Measuring instruments and measuring devices (phantoms, MR coils, etc.) may not be entered in the table if they have already been entered in the mobile device.

Measuring equipment / -instruments	Serial No.	Date Used

2 General Tests

2.1 System

SI Visual inspection of system

☐ ☐ ☐

SI Inspecting visible cables and cable guides

☐ ☐ ☐

2.2 Options and accessories

2.2.3 List of options and accessories

SI Head coil

☐ ☐ ☐

Material No.:	
Serial No.:	

SI Neck coil

☐ ☐ ☐

Material No.:	
Serial No.:	

SI Body matrix coil

☐ ☐ ☐

Material No.:	
Serial No.:	

SI Spine matrix coil

☐ ☐ ☐

Material No.:	
Serial No.:	

SI Extremity Coil

☐ ☐ ☐

Material No.:	
Serial No.:	

SI Breast coil

☐ ☐ ☐

Material No.:	
Serial No.:	

OK not OK n.a.

SI TX / RX head coil

☐ ☐ ☐

Material No.:	
Serial No.:	

SI Loop coil

☐ ☐ ☐

Material No.:	
Serial No.:	

SI Flex coil interface

☐ ☐ ☐

Material No.:	
Serial No.:	

SI Flex coil small

☐ ☐ ☐

Material No.:	
Serial No.:	

SI Flex coil large

☐ ☐ ☐

Material No.:	
Serial No.:	

SI Wrist coil

☐ ☐ ☐

Material No.:	
Serial No.:	

SI Foot ankle coil

☐ ☐ ☐

Material No.:	
Serial No.:	

OK not OK n.a.

SI PA matrix coil

☐ ☐ ☐

Material No.:	
Serial No.:	

SI Knee coil

☐ ☐ ☐

Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: _____

Date: Name:

2.2.3.1 Option MRSC

Option present: Yes: ☐ No: ☐

Signature: _____

Date: Name:

SI Option MRSC

☐ ☐ ☐

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: _____

Date: Name:

OK not
OK OK n.a.

2.2.3.2 Option Trolley

Option present:

Yes: ☐No: ☐

Signature: _____

Date: Name:

SI Option Trolley

☐ ☐ ☐

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: _____

Date: Name:

2.2.3.3 Option 1

Option present:

Yes: ☐No: ☐

Signature: _____

Date: Name:

SI Option 1

☐ ☐ ☐

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: _____

Date: Name:

OK not
OK OK n.a.

2.2.3.4 Option 2

Option present:

Yes: ☐No: ☐

Signature: _____

Date:

Name:

SI Option 2

☐ ☐ ☐

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: _____

Date:

Name:

2.2.3.5 Option 3

Option present:

Yes: ☐No: ☐

Signature: _____

Date:

Name:

SI Option 3

☐ ☐ ☐

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: _____

Date:

Name:

OK not
OK OK n.a.

2.2.3.6 Option 4

Option present: Yes: ☐ No: ☐

Signature: _____

Date: Name:

SI Option 4 ☐ ☐ ☐

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: _____

Date: Name:

2.2.3.7 Option 5

Option present: Yes: ☐ No: ☐

Signature: _____

Date: Name:

OK not OK n.a.

☐ ☐ ☐

SI Option 5

Description:	
Material No.:	
Serial No.:	

Work steps of this section were performed by:

Signature: _____

Date:

Name:

2.3 Phantoms

SI Checking the phantoms

☐ ☐ ☐
2.4 Quench tube**2.4.1 Quench tube inspection**

SI Visual inspection of the quench tube

☐ ☐ ☐
2.5 Magnet**2.5.1 Quench valve**

SI Check for water in the quench valve

☐ ☐ ☐
2.5.2 Ice formation check

SI Check for ice formation at the magnet service turret and venting system

☐ ☐ ☐
2.5.3 Leakage check

SI Magnet system leak check

☐ ☐ ☐
2.6 Patient fan

SI Checking the air filter of the patient fan

☐ ☐ ☐
2.7 User documentation

SI User documentation available and legible

☐ ☐ ☐
2.8 User icons, button labels

SI Checking user icons and button labels

☐ ☐ ☐

2.9 Identification of the controlled access area**2.9.1 Identification of the controlled access area (0.5 mT area)**

- SI Identification of the controlled access area (0.5 mT area)
- SI Checking the warning signs (SIEMENS)
- SI Checking the warning signs (customer signs cover all safety aspects as listed on the SIEMENS signs)

☐	☐	☐
☐	☐	☐
☐	☐	☐

2.10 RF room door**2.10.1 Door lock**

- SI Checking the door lock of the RF room

☐	☐	☐
--------------------------	--------------------------	--------------------------

2.11 Warning Signs**2.11.1 Laser warning labels**

- SI Checking the laser warning labels

☐	☐	☐
--------------------------	--------------------------	--------------------------

2.11.2 Warning signs of the Patient Table

- SI Checking the warning signs of the patient table

☐	☐	☐
--------------------------	--------------------------	--------------------------

2.11.3 Hearing protection sign

- SI Checking the hearing protection sign

☐	☐	☐
--------------------------	--------------------------	--------------------------

3 Electrical Tests**3.1 Measurement of the ground wire****3.1.1 Measurement of the system ground wire (every 2 years)**

- SI ACC - ground bolt - line distribution box (limit value $\leq 0.3 \Omega$)

☐	☐	☐
--------------------------	--------------------------	--------------------------

Value:

- SI RF-filter plate - ground bolt (limit value $\leq 0.3 \Omega$)

☐	☐	☐
--------------------------	--------------------------	--------------------------

Value:

- SI Magnet - magnet housing (limit value $\leq 0.3 \Omega$)

☐	☐	☐
--------------------------	--------------------------	--------------------------

Value:

- SI RF components on the magnet - mounting plate (limit value $\leq 0.3 \Omega$)

☐	☐	☐
--------------------------	--------------------------	--------------------------

Value:

- SI Patient table - screw at table underside (limit value $\leq 0.3 \Omega$)

☐	☐	☐
--------------------------	--------------------------	--------------------------

Value:

OK not
OK OK n.a.SI Magnet cold head compressor - housing (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Value:

SI MRC-Host - housing screw (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Value:

SI Option SEP - metal frame (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Value:

SI Option IFP - metal frame (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Value:

SI Intercom operating unit -housing screw (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Value:

SI Intercom - housing screw (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Value:

SI Power distribution for the console components - ground screw (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Value:

SI MRC TFT monitor - housing screw at the bottom (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Value:

SI Option 1 (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Description:	
Material No.:	
Serial No.:	

Value:

OK not OK n.a.

SI Option 2 (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Description:	
Material No.:	
Serial No.:	

Value:

SI Option 3 (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Description:	
Material No.:	
Serial No.:	

Value:

Work steps of this section were performed by:

Signature: _____

Date:

Name:

3.1.2 Measurement of the MRSC ground wire (annually)

Option present:

Yes: ☐No: ☐

Signature: _____

Date:

Name:

SI MRSC Host - housing screw (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Value:

SI TFT monitor housing screw at the bottom (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Value:

OK not OK n.a.

SI Option external desktop MOD (limit value $\leq 0.3 \Omega$)
☐ ☐ ☐

Value:

Work steps of this section were performed by:

Signature: _____

Date:

Name:

4 Mechanical Tests

4.1 TALES

SI Replacing TALES (every 2 years)

☐ ☐ ☐

5 Function Tests

5.1 Emergency Buttons

SI Emergency Shutdown Buttons

☐ ☐ ☐

SI Electronic cabinets are de-energized

☐ ☐ ☐

SI "Line" LED on the alarm is off

☐ ☐ ☐

SI Option UPS type Vertiv Liebert GXT-4: Display shows output voltage = 0 V

☐ ☐ ☐

SI Option UPS type Powerware 9130G-3000T-XLEU: Blue LED S1 in the electronic cabinet is off

☐ ☐ ☐

5.2 Gradient Supervision Avanto

SI Checking the gradient supervision

☐ ☐ ☐

5.3 Gradient Supervision Espree

SI Checking the gradient supervision

☐ ☐ ☐

5.4 Patient Table

5.4.1 Safety switches

SI Checking the safety switches

☐ ☐ ☐

5.4.2 STOP function

SI Checking the patient table stop function

☐ ☐ ☐

5.4.3 Emergency movement

SI Checking the patient table emergency movement

☐ ☐ ☐

OK not
OK OK n.a.**5.4.4 Patient table bruising hazard****5.4.4.1 Service end**

SI Checking the distance between the patient table and the wall in the RF-room

☐ ☐ ☐
5.4.4.2 Cover

SI Checking the distance between the patient table and the cover

☐ ☐ ☐
5.4.5 Squeeze bulb

SI Checking the squeeze bulb

☐ ☐ ☐
5.4.6 Table movement

SI Checking the patient table movement

☐ ☐ ☐
5.5 Patient Trolley**5.5.1 Option: Trolley**

SI Option trolley: Checking the locking mechanism

☐ ☐ ☐
5.6 ERDU**5.6.1 ERDU**

SI ERDU Test

☐ ☐ ☐
5.6.2 ERDU (MSUP K2306)

SI Magnet Stop button test (ERDU)

☐ ☐ ☐
5.7 RF Integrity**5.7.1 SAR Monitor Test****5.7.1.4 SAR Check**

SI Checking the SAR monitor (only for systems with SW version VB13 or earlier)

☐ ☐ ☐
5.7.2 Measurement of body Coil Power Loss

SI Measurement of the Coil Power Loss

☐ ☐ ☐
5.8 Quality Assurance (QA)

SI General Quality Assurance

☐ ☐ ☐

All documents may only be used for rendering services on Siemens Healthcare Products. Any document in electronic form may be printed once. Copy and distribution of electronic documents and hardcopies is prohibited. Offenders will be liable for damages. All other rights are reserved.

healthcare.siemens.com/services

Siemens Healthineers Headquarters
Siemens Healthcare GmbH
Henkestr. 127
91052 Erlangen
Germany
Telephone: +49 9131 84-0
siemens.com/healthineers

Print No.: M6-020.834.05.03.02 | Replaces: M6-020.834.05.02.02
Doc. Gen. Date: 11.17 | Language: English
© Siemens Healthcare GmbH, 2015

siemens.com/healthineers